Curriculum Vitae – Zirui Zhao

CONTACT

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EDUCATION

Doctor of Philosophy (Ph.D.) in Computer Science — *Aug 2020 -*
National University of Singapore
Advisor: Prof. Lee Wee Sun

Bachelor of Engineering (B.Eng.) in Automation (EECS) — *Aug 2016 - Jun 2020*
Xi'an Jiaotong University — GPA: 3.92/4.30
Elite Qian Xuesen Program, excellent graduate.

VISITING

Carnegie Mellon University, Pittsburgh, U.S.A. — *Jul 2019 - Aug 2019*
Summer Intern, Safe AI Lab
The Robotics Institute & Dept. Mechanical Engineering.
Advisor: Prof. Ding Zhao

National University of Singapore, Singapore — *Jul 2018 - Aug 2018*
Visiting Student, 2018 Summer Workshop
School of Computing.
Advisor: Prof. Soo Yuen Jien

INTERESTS

Robotics; Robot Decision-making under uncertainty; Robot Planning with Large Language Models; Robot Perception; Human-robot interactions.

SKILLS

Languages: Python, C++, MATLAB, LaTeX.
Application Tools: TensorFlow, PyTorch, Keras, Scikit-Learn, Numpy, Jupyter, OpenCV, ROS.
Software: Carla, GitLab, GitHub.

RESEARCH

Planning with Large Language Models — *2023 -*
Supervisor: Prof. Wee Sun Lee and Prof. David Hsu — *NUS*
We use Large Language Models as both the commonsense model of the world and the heuristic policy within the Monte Carlo Tree Search framework, enabling practical and reasoned decision-making for daily tasks.
- LLM's world model provides a commonsense prior belief of states for MCTS to achieve reasoned decision-making efficiently.
- The LLM's heuristic policy guides the search to relevant parts of the tree, substantially reducing the search complexity.

Grounding Natural Language Instructions for Object Placing — *2021 -*
Supervisor: Prof. Wee Sun Lee and Prof. David Hsu — *NUS*
We proposed a novel method, ParaGon, for language-conditioned object placing.
ParaGon integrates a parsing algorithm into an end-to-end trainable neural network. It is data-efficient and generalizable for learning compositional instructions, and robust to noisy, ambiguous language inputs.

Active Risk-sensitive Inverse Reinforcement Learning — *Jul 2019 - Aug 2019*
Supervisor: Ding Zhao — *Carneige Mellon University*
- Active demonstration querying for faster human risk envelope approximation via disturbance planning.
- Experimental verification in single-step and multi-step setting with simulated car-following task in Carla.

Visual Semantic SLAM for Outdoor Environments *Nov 2018 - Jun 2019*
Supervisor: Pengju Ren, Cognitive Architecture Group *Xi'an Jiaotong University*
- Accomplished visual semantic SLAM based on PSPNet101 and ORB SLAM.
- The SLAM has implemented with GPS Fusion and topological semantic mapping.
- GitHub: `https://github.com/1989Ryan/Semantic_SLAM`

TEACHING

CS 3263 Foundations of Artificial Intelligence *Sem 2 AY22/23 & Sem 1 AY23/24*
Instructor: Prof. Tze-Yun Leong *NUS*
- Tutor for course tutorials.
- Teach topics that cover Logic, Advanced Search, Constraint Satisfaction, Bayesian Networks, Causal Networks, Markov Decision Process, and Reinforcement Learning.

AWARDS

- **Best Paper Runner-up** LTAMP workshop @ Robotics: Science & Systems, 2023
- **NUS Research Scholarship** National University of Singapore, 2020 - 2024
- **Excellant Graduate** Xi'an Jiaotong University, 2020
- **Dean's List** Xi'an Jiaotong University, 2017 & 2018 & 2019
- **2-nd of 1989 Mechanical Alumni Scholarship** Xi'an Jiaotong University, 2018
- **Siyuan Merit Scholarship** Xi'an Jiaotong University, 2017

PUBLICATION

[4] **Z. Zhao**, W. S. Lee, and D. Hsu. "Large Language Models as Commonsense Knowledge for Large-Scale Task Planning". In: *Thirty-seventh Conference on Neural Information Processing Systems (NeurIPS)*. 2023.

[3] **Z. Zhao**, W. S. Lee, and D. Hsu. "Differentiable Parsing and Visual Grounding of Natural Language Instructions for Object Placement". In: *2023 IEEE International Conference on Robotics and Automation (ICRA)*. 2023, pp. 11546–11553. DOI: `10.1109/ICRA48891.2023.10160640`.

[2] R. Chen, W. Wang, **Z. Zhao**, and D. Zhao. "Active learning for risk-sensitive inverse reinforcement learning". In: *arXiv preprint arXiv:1909.07843* (2019).

[1] **Z. Zhao**, Y. Mao, Y. Ding, P. Ren, and N. Zheng. "Visual-based semantic SLAM with landmarks for large-scale outdoor environment". In: *2019 2nd China Symposium on Cognitive Computing and Hybrid Intelligence (CCHI)*. IEEE. 2019, pp. 149–154.